

Accuracy-Preserving Stability Regularization for Large-Scale Retail Demand Forecasting

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Abstract

Retail demand forecasts are reused across replenishment, capacity, labor, and transportation planning cycles. Training objectives based on prediction error do not limit sharp changes between forecasts for consecutive periods. Post hoc smoothing adjusts forecasts after training. We test whether penalizing consecutive within-series forecast changes during training improves stability without substantially changing point accuracy. The model combines recent demand embeddings with calendar, price, hierarchy, item, and store features. On selected M5 subsets of 1000, 3000, and 4000 series, the stability aware hybrid model improves Forecast Stability Score over XGBoost by 6.91%, 6.66%, and 7.68%. RMSE changes remain within 0.72% across three random seeds. Post hoc exponential smoothing reduces raw forecast movement more but causes a larger increase in RMSE. Regularization during training keeps RMSE closer to the XGBoost baseline. It also gives lower normalized FSS than post hoc smoothing. These results support evaluating both accuracy and stability in retail forecasting.

CCS Concepts • Computing methodologies • Machine learning • Machine Learning Algorithms • Regularization

Additional Keywords and Phrases: Retail demand forecasting; forecast stability; supply chain management; business intelligence; XGBoost; time-series forecasting; stability regularization; e-commerce analytics

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1 INTRODUCTION

Demand forecasts guide replenishment, inventory, production, labor, and transportation planning [1, 2, 3]. Sharp changes between forecasts for consecutive periods can lead to planning adjustments, even when average forecast error is acceptable. This issue is linked to forecast stability and demand amplification [4, 5, 6]. It is especially relevant to intermittent retail demand, where periods of zero demand and occasional spikes occur alongside promotions and price changes [7, 8, 9].

Most forecasting systems are chosen based on how closely their predictions match actual demand. However, these accuracy measures do not limit changes between forecasts for consecutive periods [3, 6]. Models that use only demand history may miss structured retail information. Tabular models using lagged values and rolling statistics may not learn compact representations of recent demand patterns. Post hoc smoothing reduces movement only after fitting and may suppress responsive variation. M5 makes these tradeoffs testable by pairing item store histories with calendar and sell price tables [9]. We therefore use a lightweight multiscale temporal encoder with XGBoost for heterogeneous operational features [15].

This study tests whether regularization during training improves horizontal forecast stability without substantially reducing point accuracy. It also compares this approach with post hoc smoothing in terms of accuracy and stability. The hybrid pipeline provides a test bed for evaluating stability regularization with structured retail features.

The paper makes three contributions. First, the study adds a total variation penalty to the XGBoost training objective to limit changes between consecutive forecasts within each demand series. It compares this approach with post hoc smoothing. Second, it uses a temporal structured retail pipeline that combines a compact demand history embedding with calendar, price, hierarchy, item, and store features. Third, it evaluates point accuracy, raw and normalized stability, smoothing baselines, seed robustness, leakage, and runtime on consistent M5 subsets.

2 RELATED WORK

2.1 Deep Time Series Forecasting

Recent deep time series models improve temporal representation and point accuracy through multiscale mixing, patching, efficient attention, decomposition, and MLP based mixing [10, 11, 12, 13, 14]. They generally do not treat adjacent forecast path movement as an explicit operational objective.

Retail forecasting also depends on calendar, price, event, product, and store signals [2, 9]. Standard accuracy oriented training does not encode a preference for restrained adjacent movement [3, 6]. Temporal representation learning supports the analysis, while operational stability remains the main object of study.

2.2 Supply Chain and Retail Demand Forecasting

Retail forecasting involves sparse SKU store demand, seasonality, prices, promotions, and hierarchy. Competitions shape empirical practice [16], and M5 provides Walmart sales, calendar, and price data at scale [9]. Methods must use operational features and be evaluated in planning contexts [1, 2, 3].

Tree models handle nonlinear, heterogeneous demand drivers [2, 9]. XGBoost handles sparse features and interactions at scale [15], motivating hybrids that combine learned demand history embeddings with tabular features.

2.3 Hybrid Forecasting Frameworks

The pipeline combines a lightweight multiscale MLP inspired by TimeMixer [10] with XGBoost [15] using M5 retail features [9]. The encoder is not a full TimeMixer reproduction. The pipeline serves only as a test bed for comparing the same learner with and without stability regularization.

2.4 Forecast Stability and Robust Forecasting

Horizontal stability concerns forecasts for adjacent target dates, while vertical stability concerns revisions for the same target across forecast origins [6]. Van Belle et al. use a composite neural loss to improve rolling origin stability [22], and Caljon et al. study dynamic weighting of this loss [23]. Other studies link forecast revisions and method choice to demand amplification [4, 5, 17], and intermittent demand forecasts to inventory performance [7, 8, 18].

We examine stability regularization in a boosted-tree retail pipeline, comparing it with post-hoc smoothing and evaluating scale-normalized stability.

3 METHODOLOGY

3.1 Problem Formulation

Let $y_{\{i,t\}}$ be demand for item store series i at time t . Given historical demand $y_{\{i,t-L:t-1\}}$ and structured covariates $x_{\{i,t\}}$ containing calendar, price, event, category, and lag derived information, the task is to estimate $\hat{y}_{\{i,t\}}$.

3.2 Temporal Representation Learning and Feature Fusion

The temporal encoder maps historical demand sequences to compact embeddings:

$$z_{\{i,t\}} = g_{\theta}(y_{\{i,t-L:t-1\}}) \quad (1)$$

where $g_{\theta}(\cdot)$ is a lightweight multiscale MLP feature extractor inspired by TimeMixer [10], not a full reproduction. It transforms a shifted 28 day history using raw lags, first differences, and pooled means and standard deviations at 2, 4, 7, and 14 day resolutions. The final hidden layer yields a 16 dimensional embedding $z_{\{i,t\}}$.

The forecasting model concatenates temporal embeddings with lag, rolling, calendar, price, hierarchy, item, and store features:

$$h_{\{i,t\}} = [z_{\{i,t\}}; x_{\{i,t\}}] \quad (2)$$

and passes the fused representation to a gradient boosted forecasting layer [15]:

$$\hat{y}_{\{i,t\}} = f_{\varphi}(h_{\{i,t\}}) \quad (3)$$

The encoder's scaler and MLP are fitted only on a seeded sample of up to 50,000 training windows to limit computation. The fitted encoder generates embeddings for all rows, and validation uses the same transformations without refitting. Lag and rolling features use shifted demand, while the boosted tree models use all selected training rows.

3.3 Stability Aware Objective

The forecast loss is mean squared error:

$$L_{\text{forecast}} = (1/N) \sum_{\{i,t\}} (y_{\{i,t\}} - \hat{y}_{\{i,t\}})^2 \quad (4)$$

The stability loss penalizes consecutive forecast changes within each demand series:

$$L_{\text{stability}} = (1/N_s) \sum_i \sum_{\{t>1\}} |\hat{y}_{\{i,t\}} - \hat{y}_{\{i,t-1\}}| \quad (5)$$

where N_s is the total number of valid pairs of consecutive predictions within individual series. Equation (5) applies a discrete total variation penalty separately to each series, linking consecutive dates but not different items or stores. The total objective is

$$L_{\text{total}} = L_{\text{forecast}} + \lambda L_{\text{stability}} \quad (6)$$

where λ controls the stability accuracy trade off. We evaluate $\lambda \in \{0, 0.01, 0.05, 0.1, 0.5\}$ and use $\lambda = 0.05$ for the final model.

3.4 Implementation of the Stability Aware XGBoost Objective

The XGBoost custom objective uses row level gradients and Hessians. The MSE term contributes gradient $2(\hat{y}_{\{i,t\}} - y_{\{i,t\}})$ and Hessian 2. Stability pairs are formed after sorting training rows by item-store series and day index, using only consecutive rows from the same series.

For each consecutive pair, the later prediction receives a sign subgradient and the earlier one receives its negative. The first and last predictions in each series receive only one contribution. The sign term uses $\max(|\Delta|, \epsilon)$, and a small $\lambda\epsilon$ term keeps the Hessian positive. Validation and test targets are excluded. The penalty is applied during training and links only consecutive predictions within the same series.

3.5 Post Hoc Smoothing Baseline

To separate training time stability regularization from post hoc smoothing, we also evaluate exponential smoothing applied after prediction, following the broad use of exponential smoothing ideas in forecasting and robust forecasting [19, 3]:

$$\tilde{y}_{\{i,t\}} = \alpha \hat{y}_{\{i,t\}} + (1 - \alpha) \tilde{y}_{\{i,t-1\}} \quad (7)$$

with $\alpha \in \{0.5, 0.7, 0.9\}$. Smoothing is applied independently within each demand series and only to model predictions. It does not change the fitted forecasting model or use validation targets.

3.6 Evaluation Metrics

We report MAE, RMSE, WAPE, FSS, normalized FSS, volatility, runtime, and relative changes against XGBoost. Following forecasting metric guidance [20, 21, 3], WAPE is

$$\text{WAPE} = 100 \times (\sum_{i,t} |y_{i,t} - \hat{y}_{i,t}|) / (\sum_{i,t} |y_{i,t}|) \quad (8)$$

FSS is defined as

$$\text{FSS} = (1/N_s) \sum_i \sum_{t>1} |\hat{y}_{i,t} - \hat{y}_{i,t-1}| \quad (9)$$

Lower FSS indicates smoother consecutive forecasts. This score measures horizontal stability by comparing forecasts for consecutive target dates within the same horizon [6]. Vertical instability measures how forecasts for the same target date change across different forecast origins [22, 23]. We also report a scale normalized score:

$$\text{nFSS}_i = [(1/(T_i-1)) \sum_{t>1} |\hat{y}_{i,t} - \hat{y}_{i,t-1}|] / [(1/T_i) \sum_t |y_{i,t}| + \varepsilon] \quad (10)$$

$$\text{nFSS} = (1/M) \sum_{i=1}^M \text{nFSS}_i \quad (11)$$

where $\varepsilon = 10^{-6}$. Low volume series can still yield large nFSS values. The results therefore concern within horizon path smoothness, not rolling origin revision stability.

4 EXPERIMENTS

4.1 Dataset and Setup

We use the M5 dataset, which contains daily Walmart demand by item and store, calendar information, and selling prices [9]. Experiments cover 1000, 3000, and 4000 series. At each scale, the boosted tree setup uses all selected training rows, 300 XGBoost trees, the same features, fixed temporal encoder settings, and $\lambda = 0.05$.

All training dates come before validation dates. Features include shifted lag and rolling demand values, calendar information, selling prices, and categorical item and store variables. We repeat the experiments with seeds 42, 123, and 2024.

4.2 Error Metrics for Intermittent Demand

M5 contains many observations with zero or very low demand [9]. MAPE is undefined at zero demand and unstable near zero [3, 8, 21]. We therefore report WAPE instead.

4.3 Baselines

We compare XGBoost, the encoder alone, and two hybrid models. Both hybrids combine the encoder with XGBoost. One also includes stability regularization. The tables and figures label the encoder as “TimeMixer” or “TimeMixer only.” This is the lightweight encoder described in Section 3.2. It is not a full TimeMixer reproduction. “Hybrid + Stable” refers to the hybrid with stability regularization. The post hoc smoothing baselines (“XGBoost + Smooth” and “Hybrid + Smooth”) use $\alpha \in \{0.5, 0.7, 0.9\}$.

4.4 Main Results

Table 1 reports mean $\pm$ standard deviation across seeds, and Figure 1 summarizes RMSE and FSS. XGBoost remains the strongest baseline for point accuracy. The stability aware hybrid model achieves the lowest FSS among the primary training methods. These results show a tradeoff between accuracy and stability, not superiority in RMSE.

Table 1. Mean and standard deviation across three random seeds under consistent full training configurations. WAPE is reported as the percentage error metric for intermittent retail demand.

Series	Method	Runs	MAE	RMSE	WAPE (%)	FSS	Volatility
1000	XGBoost	3	0.926 $\pm$ 0.034	1.774 $\pm$ 0.185	70.02 $\pm$ 4.82	0.220 $\pm$ 0.028	0.313 $\pm$ 0.043
1000	TimeMixer	3	0.970 $\pm$ 0.044	1.930 $\pm$ 0.316	73.28 $\pm$ 4.30	0.347 $\pm$ 0.076	0.391 $\pm$ 0.073
1000	Hybrid	3	0.926 $\pm$ 0.034	1.781 $\pm$ 0.198	70.02 $\pm$ 4.76	0.222 $\pm$ 0.030	0.315 $\pm$ 0.044
1000	Hybrid + Stable	3	0.927 $\pm$ 0.035	1.787 $\pm$ 0.208	70.08 $\pm$ 4.71	0.205 $\pm$ 0.030	0.302 $\pm$ 0.042
3000	XGBoost	3	0.953 $\pm$ 0.006	1.852 $\pm$ 0.081	69.81 $\pm$ 1.47	0.228 $\pm$ 0.007	0.326 $\pm$ 0.012
3000	TimeMixer	3	0.993 $\pm$ 0.012	2.002 $\pm$ 0.123	72.79 $\pm$ 1.37	0.401 $\pm$ 0.040	0.439 $\pm$ 0.027
3000	Hybrid	3	0.953 $\pm$ 0.007	1.850 $\pm$ 0.078	69.84 $\pm$ 1.43	0.229 $\pm$ 0.008	0.328 $\pm$ 0.013
3000	Hybrid + Stable	3	0.954 $\pm$ 0.007	1.854 $\pm$ 0.082	69.91 $\pm$ 1.45	0.213 $\pm$ 0.009	0.315 $\pm$ 0.012
4000	XGBoost	3	0.955 $\pm$ 0.013	1.876 $\pm$ 0.012	69.53 $\pm$ 1.00	0.232 $\pm$ 0.007	0.331 $\pm$ 0.006
4000	TimeMixer	3	0.998 $\pm$ 0.020	2.014 $\pm$ 0.037	72.65 $\pm$ 1.34	0.350 $\pm$ 0.027	0.399 $\pm$ 0.031
4000	Hybrid	3	0.956 $\pm$ 0.012	1.882 $\pm$ 0.012	69.56 $\pm$ 0.94	0.231 $\pm$ 0.005	0.330 $\pm$ 0.005
4000	Hybrid + Stable	3	0.956 $\pm$ 0.012	1.881 $\pm$ 0.014	69.58 $\pm$ 0.96	0.214 $\pm$ 0.006	0.318 $\pm$ 0.007

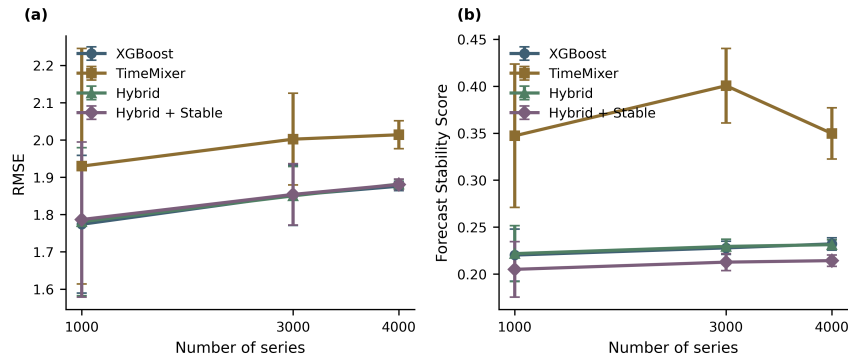


Figure 1. Main results under consistent full training configurations. Error bars denote standard deviation across three random seeds.

4.5 Stability Accuracy Tradeoff

The results are consistent across all three scales. Table 2 shows that the stability aware hybrid model improves FSS over XGBoost by 6.91%, 6.66%, and 7.68% at 1000, 3000, and 4000 series. The largest absolute RMSE change is 0.72%. Figure 2 compares accuracy and stability across methods. Regularization during training produces smoother forecasts, and RMSE remains close to the structured baseline.

Table 2. Relative stability, accuracy, and volatility changes for the stability aware hybrid model versus XGBoost.

Series	FSS Improvement (%)	RMSE Change (%)	Volatility Improvement (%)	Seeds Improved	Runs
1000	6.91	0.72	3.77	3	3
3000	6.66	0.10	3.18	3	3
4000	7.68	0.24	3.85	3	3

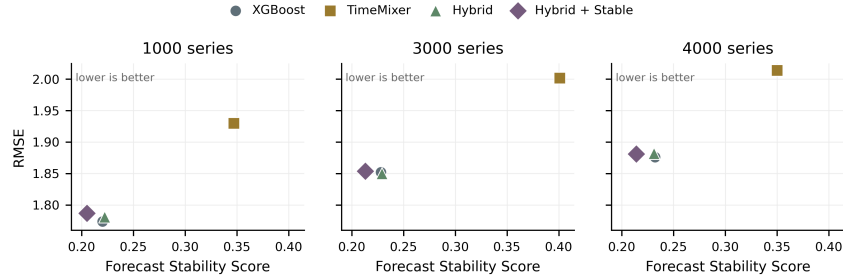


Figure 2. Accuracy stability tradeoff under the revised consistent scalability setup. Points show method level operating positions at each scale.

4.6 Smoothing Baselines and Normalized Stability

Table 3 compares training time regularization with the strongest post hoc smoothing setting from the tested grid. The comparison reveals a clear tradeoff. With $\alpha = 0.5$, exponential smoothing achieves the lowest raw FSS. It reduces changes between consecutive predictions but increases RMSE. Compared with XGBoost, Hybrid + Smooth increases RMSE by about 1.66% to 2.33%. The RMSE change for Hybrid + Stable remains within 0.72%.

The same pattern appears in the accuracy stability tradeoff: post hoc smoothing reduces raw movement more aggressively, while the stability aware hybrid model preserves more accuracy oriented forecast structure.

Table 3. Post hoc smoothing baseline comparison. Smoothing rows use the best raw FSS setting from $\alpha \in \{0.5, 0.7, 0.9\}$; lower FSS and nFSS are better.

Series	Method	α	RMSE	WAPE (%)	FSS	nFSS	RMSE Δ (%)
1000	Hybrid + Stable	—	1.787 $\pm$ 0.208	70.08 $\pm$ 4.71	0.205 $\pm$ 0.030	388.7 $\pm$ 113.8	0.72
1000	XGBoost + Smooth	0.5	1.812 $\pm$ 0.217	70.54 $\pm$ 4.56	0.116 $\pm$ 0.015	449.0 $\pm$ 132.9	2.15
1000	Hybrid + Smooth	0.5	1.815 $\pm$ 0.223	70.53 $\pm$ 4.55	0.116 $\pm$ 0.015	423.4 $\pm$ 94.7	2.33
3000	Hybrid + Stable	—	1.854 $\pm$ 0.082	69.91 $\pm$ 1.45	0.213 $\pm$ 0.009	373.5 $\pm$ 56.1	0.10
3000	XGBoost + Smooth	0.5	1.887 $\pm$ 0.090	70.41 $\pm$ 1.43	0.118 $\pm$ 0.004	452.1 $\pm$ 90.3	1.87
3000	Hybrid + Smooth	0.5	1.883 $\pm$ 0.087	70.41 $\pm$ 1.39	0.119 $\pm$ 0.004	437.3 $\pm$ 50.0	1.66
4000	Hybrid + Stable	—	1.881 $\pm$ 0.014	69.58 $\pm$ 0.96	0.214 $\pm$ 0.006	377.5 $\pm$ 73.7	0.24
4000	XGBoost + Smooth	0.5	1.911 $\pm$ 0.016	70.12 $\pm$ 0.92	0.121 $\pm$ 0.004	416.9 $\pm$ 74.6	1.88
4000	Hybrid + Smooth	0.5	1.917 $\pm$ 0.021	70.15 $\pm$ 0.88	0.120 $\pm$ 0.003	423.3 $\pm$ 90.1	2.16

Scale normalized stability supports the same interpretation: the stability aware hybrid model improves nFSS over XGBoost by 40.21%, 45.66%, and 41.93% at 1000, 3000, and 4000 series and outperforms the best Hybrid

+ Smooth baseline on nFSS. Because nFSS divides by average realized demand, we interpret it as a relative metric rather than an operational threshold.

4.7 Scalability and Runtime Analysis

Table 4 reports runtime under consistent full row, 300 tree configurations. Relative to XGBoost, the stability aware hybrid model incurs multipliers of 4.39 $\times$, 7.71 $\times$, and 9.72 $\times$ at 1000, 3000, and 4000 series. Post hoc smoothing adds little processing time after prediction but does not reduce the cost of fitting the underlying models.

Table 4. Runtime scalability under the revised consistent configuration. All main scales use full selected training rows and 300 XGBoost trees.

Series	Method	Runtime Mean (s)	Runtime Std (s)	Peak Memory (MB)
1000	XGBoost	17.31	6.53	3147.36
1000	TimeMixer	27.24	4.28	3147.36
1000	Hybrid	47.91	9.83	3147.36
1000	Hybrid + Stable	75.90	13.97	3147.36
3000	XGBoost	45.77	4.61	2893.42
3000	TimeMixer	187.26	21.47	2893.42
3000	Hybrid	269.29	38.70	2893.42
3000	Hybrid + Stable	352.97	39.81	2893.42
4000	XGBoost	53.09	10.63	3352.53
4000	TimeMixer	263.15	41.01	3352.53
4000	Hybrid	372.59	73.37	3352.53
4000	Hybrid + Stable	515.96	61.28	3352.53

4.8 Robustness Analysis

Across the three random seeds, the stability aware hybrid model improves FSS over XGBoost at all three evaluated scales. The average FSS improvement across 1000, 3000, and 4000 series is 7.08%, with a maximum absolute RMSE change of 0.72%.

5 DISCUSSION

In these experiments, post hoc exponential smoothing is inexpensive to apply and achieves the lowest raw FSS. Regularization during training reduces raw movement less, but better preserves point accuracy and achieves lower scale normalized FSS.

Forecast changes can affect replenishment and upstream planning [4, 5, 17]. Planners may therefore prefer moderate stabilization over stronger smoothing when accuracy and responsiveness to demand are important.

5.1 Implications for Forecast Evaluation

Point accuracy alone is not enough for repeated planning. Models with similar RMSE can produce different levels of forecast movement. Horizontal stability measures changes between forecasts for consecutive target

dates. It differs from point error and revisions for the same target across forecast origins [6, 22, 23]. Regularization during training and post hoc smoothing offer different balances between accuracy and stability. Smoothing can reduce raw movement more. Regularization can better preserve point accuracy. These tradeoffs can help planners choose a model. However, the experiments evaluate forecasts, not inventory outcomes. They do not establish inventory savings.

The runtime cost is real: temporal embedding extraction and stability aware boosted tree training make the final method slower than XGBoost, with multipliers of 4.39×, 7.71×, and 9.72×. These values reflect end to end experimental cost, not deployment time inference latency.

Many supply chain forecasts are generated in scheduled batches rather than in real time. Longer training times may be acceptable if more stable forecasts reduce downstream planning disruptions. This study measures total experimental runtime. It does not separate training time from inference time. Future work should measure inference time using realistic production batch sizes. This should include temporal embedding extraction and XGBoost prediction.

6 LIMITATIONS

The evaluation is limited to selected M5 subsets rather than the full hierarchy. M5 includes sparse demand by item and store, prices, events, and hierarchy. It does not cover every supply chain setting. The largest experiment under the full configuration uses 4000 series.

FSS and nFSS measure horizontal movement across adjacent target dates, not vertical forecast revisions across rolling origins. Low volume intermittent series can also yield large nFSS values, so the score is interpreted comparatively rather than as an operational threshold.

Finally, the evidence remains forecast level rather than inventory level. No replenishment simulation establishes changes in service levels, stockouts, holding costs, or schedule revisions. These outcomes, together with rolling origin evaluation and broader retail datasets, are priorities for future work.

7 CONCLUSION

This paper examined whether horizontal forecast path stability can be regularized during training without materially changing point accuracy. Across 1000, 3000, and 4000 M5 series, the same series movement penalty improves FSS while keeping RMSE changes small. Post hoc smoothing is preferable when raw smoothness is the only goal. Regularization during training better preserves accuracy and performs well on scale normalized stability. However, the study uses only one dataset and one held out horizon and evaluates forecasts rather than inventory outcomes.

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